Energy-Normalized RKHS Pearson Classification: Threshold and Separation Guarantees

1 Pearson discrepancy and energy normalization

Let $\mathcal{H}$ be a real separable RKHS with measurable feature map ϕ. For a reference distribution P_i, define

$$\mu_i = \mathbb{E}_{P_i}\phi(X), \qquad T_i = \mathbb{E}_{P_i}[\phi(X)\otimes\phi(X)], \qquad A_i = T_i + \lambda I,$$

where $\lambda > 0$ and the outer-product operator is defined by

$$(u\otimes v)f = \langle v, f\rangle_{\mathcal{H}}\, u.$$

For an incoming distribution Q, write

$$\mu_Q = \mathbb{E}_Q\phi(Z), \qquad \Delta_i = \mu_Q - \mu_i.$$

We use the regularized RKHS Pearson-type discrepancy

$$D_i(Q) = \frac{1}{2}\left\langle \Delta_i, A_i^{-1}\Delta_i\right\rangle_{\mathcal{H}}. \tag{1}$$

Its connection to Pearson is the quadratic variational form

$$D_i(Q) = \sup_{f\in\mathcal{H}}\left\{\mathbb{E}_Q f - \mathbb{E}_{P_i} f - \frac{1}{2}\left(\mathbb{E}_{P_i} f^2 + \lambda\|f\|_{\mathcal{H}}^2\right)\right\}. \tag{2}$$

Indeed, the expression being maximized equals

$$\langle f, \Delta_i\rangle_{\mathcal{H}} - \frac{1}{2}\langle f, A_i f\rangle_{\mathcal{H}} = D_i(Q) - \frac{1}{2}\left\|A_i^{1/2}(f - A_i^{-1}\Delta_i)\right\|_{\mathcal{H}}^2.$$

Let

$$v_i = A_i^{-1/2}\mu_Q, \qquad w_i = A_i^{-1/2}\mu_i.$$

Then

$$D_i(Q) = \frac{1}{2}\|v_i - w_i\|_{\mathcal{H}}^2.$$

Define the total energy and normalized score by

$$E_i(Q) = \left\langle \mu_Q, A_i^{-1}\mu_Q\right\rangle_{\mathcal{H}} + \left\langle \mu_i, A_i^{-1}\mu_i\right\rangle_{\mathcal{H}}, \tag{3}$$

$$S_i(Q) = \frac{D_i(Q)}{E_i(Q)} = \frac{\|v_i - w_i\|_{\mathcal{H}}^2}{2(\|v_i\|_{\mathcal{H}}^2 + \|w_i\|_{\mathcal{H}}^2)}. \tag{4}$$

Thus, normalization preserves the reference second-moment geometry and measures discrepancy relative to total embedding energy. It defines a normalized Pearson-type score, not the classical Pearson divergence.

If $E_i(Q) = 0$, define $S_i(Q) = 0$. Since

$$\|v_i - w_i\|_{\mathcal{H}}^2 \le 2(\|v_i\|_{\mathcal{H}}^2 + \|w_i\|_{\mathcal{H}}^2),$$

we have $0 \le S_i(Q) \le 1$.

2 Assumptions and empirical quantities

Assumption 1 (Kernel and sampling)**.** *The positive-semidefinite kernel is fixed before sampling and satisfies*

$$k(x,y) \geq 0, \qquad k(x,x) \leq K$$

for all x, y, where $K > 0$. The two windows are independent and satisfy

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_i, \qquad Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} Q.$$

These kernel assumptions imply

$$\|\phi(x)\|_{\mathcal{H}} \leq \sqrt{K}, \qquad \|\phi(x) - \phi(y)\|_{\mathcal{H}} \leq \sqrt{2K}. \tag{5}$$

Indeed,

$$\|\phi(x) - \phi(y)\|_{\mathcal{H}}^2 = k(x,x) + k(y,y) - 2k(x,y) \leq 2K.$$

The second bound uses pointwise nonnegativity of the kernel.

Define

$$\widehat{\mu}_i = \frac{1}{n}\sum_{j=1}^{n} \phi(X_j), \qquad \widehat{\mu}_Q = \frac{1}{n}\sum_{j=1}^{n} \phi(Z_j),$$

and

$$\widehat{T}_i = \frac{1}{n}\sum_{j=1}^{n} \phi(X_j) \otimes \phi(X_j), \qquad \widehat{A}_i = \widehat{T}_i + \lambda I.$$

The empirical quantities are

$$\widehat{D}_i = \frac{1}{2}\left\langle \widehat{\mu}_Q - \widehat{\mu}_i, \widehat{A}_i^{-1}(\widehat{\mu}_Q - \widehat{\mu}_i) \right\rangle_{\mathcal{H}}, \tag{6}$$

$$\widehat{E}_i = \left\langle \widehat{\mu}_Q, \widehat{A}_i^{-1}\widehat{\mu}_Q \right\rangle_{\mathcal{H}} + \left\langle \widehat{\mu}_i, \widehat{A}_i^{-1}\widehat{\mu}_i \right\rangle_{\mathcal{H}}, \tag{7}$$

$$\widehat{S}_i = \frac{\widehat{D}_i}{\widehat{E}_i}. \tag{8}$$

Set $\widehat{S}_i = 0$ if $\widehat{E}_i = 0$. The same squared-norm inequality as above gives $0 \leq \widehat{S}_i \leq 1$.

The incoming window is accepted as compatible with reference i when $\widehat{S}_i \leq \eta$.

Assumption 2 (Reference embedding lower bound)**.** *A known constant $m_0 > 0$ satisfies*

$$\|\mu_i\|_{\mathcal{H}} \geq m_0.$$

This assumption controls the normalization denominator under the null. It is separate from the Pearson separation assumption introduced below.

3 Deterministic energy bounds

3.1 Basic operator inequalities

For every $f \in \mathcal{H}$, Cauchy–Schwarz gives

$$\begin{aligned}
\left\langle f, \widehat{T}_i f \right\rangle_{\mathcal{H}} &= \frac{1}{n}\sum_{j=1}^{n} |\, \langle f, \phi(X_j)\rangle_{\mathcal{H}} \,|^2 \\
&\leq \frac{1}{n}\sum_{j=1}^{n} \|f\|_{\mathcal{H}}^2 \, \|\phi(X_j)\|_{\mathcal{H}}^2 \\
&\leq K \, \|f\|_{\mathcal{H}}^2 .
\end{aligned}$$

Hence

$$0 \preceq \widehat{T}_i \preceq KI, \qquad \lambda I \preceq \widehat{A}_i \preceq (K+\lambda)I.$$

Inversion reverses the operator order:

$$\frac{1}{K+\lambda} I \preceq \widehat{A}_i^{-1} \preceq \frac{1}{\lambda} I. \tag{9}$$

The same inequalities hold for T_i and A_i.

To justify the inverse-order rule, suppose $A \succeq C$, where both operators are bounded, self-adjoint, and bounded below by a positive multiple of I. Then

$$C^{-1/2} A C^{-1/2} \succeq I,$$

and consequently

$$A^{-1} = C^{-1/2}(C^{-1/2} A C^{-1/2})^{-1} C^{-1/2} \preceq C^{-1}.$$

This argument remains valid in an infinite-dimensional RKHS.

3.2 Energy lower bound

Lemma 1 (Midpoint decomposition)**.** *Let*

$$\widehat{c}_i = \frac{\widehat{\mu}_Q + \widehat{\mu}_i}{2}.$$

Then

$$\widehat{E}_i = \widehat{D}_i + 2\left\langle \widehat{c}_i, \widehat{A}_i^{-1}\widehat{c}_i \right\rangle_{\mathcal{H}}, \tag{10}$$

$$\widehat{E}_i \geq \widehat{D}_i + \frac{2\,\|\widehat{c}_i\|_{\mathcal{H}}^2}{K+\lambda}. \tag{11}$$

Proof. Write $x = \widehat{\mu}_Q$, $y = \widehat{\mu}_i$, and $B = \widehat{A}_i^{-1}$. Since B is self-adjoint,

$$\widehat{D}_i = \frac{1}{2}\langle x, Bx\rangle_{\mathcal{H}} + \frac{1}{2}\langle y, By\rangle_{\mathcal{H}} - \langle x, By\rangle_{\mathcal{H}}.$$

On the other hand,

$$\begin{aligned} 2\,\langle \widehat{c}_i, B\widehat{c}_i\rangle_{\mathcal{H}} &= \frac{1}{2}\langle x+y, B(x+y)\rangle_{\mathcal{H}} \\ &= \frac{1}{2}\langle x, Bx\rangle_{\mathcal{H}} + \frac{1}{2}\langle y, By\rangle_{\mathcal{H}} + \langle x, By\rangle_{\mathcal{H}}. \end{aligned}$$

Adding cancels the cross terms and proves (10).

Using (9),

$$\langle \widehat{c}_i, B\widehat{c}_i\rangle_{\mathcal{H}} \geq \frac{\|\widehat{c}_i\|_{\mathcal{H}}^2}{K+\lambda},$$

which proves (11). □

3.3 Energy upper bound

Lemma 2 (Uniform energy upper bound)**.** *For every incoming distribution Q,*

$$E_i(Q) \leq M_\lambda, \qquad \widehat{E}_i \leq M_\lambda,$$

where

$$M_\lambda = \frac{K}{\lambda} + \frac{K}{K+\lambda}. \tag{12}$$

Proof. **Step 1: Bound the incoming energy.** Every empirical mean embedding satisfies

$$\left\| \frac{1}{n} \sum_j \phi(X_j) \right\|_{\mathcal{H}} \leq \frac{1}{n} \sum_j \|\phi(X_j)\|_{\mathcal{H}} \leq \sqrt{K}.$$

Consequently,

$$\left\langle \widehat{\mu}_Q, \widehat{A}_i^{-1} \widehat{\mu}_Q \right\rangle_{\mathcal{H}} \leq \frac{\|\widehat{\mu}_Q\|_{\mathcal{H}}^2}{\lambda} \leq \frac{K}{\lambda}.$$

Step 2: Use the reference covariance identity. Write $y = \widehat{\mu}_i$. Expanding the empirical covariance gives

$$\widehat{T}_i - y \otimes y = \frac{1}{n} \sum_j (\phi(X_j) - y) \otimes (\phi(X_j) - y) \succeq 0.$$

Therefore,

$$\widehat{A}_i \succeq \lambda I + y \otimes y, \qquad \widehat{A}_i^{-1} \preceq (\lambda I + y \otimes y)^{-1}.$$

Step 3: Evaluate the rank-one inverse along y. By definition,

$$(y \otimes y)y = \|y\|_{\mathcal{H}}^2 \, y.$$

It follows that

$$(\lambda I + y \otimes y) \left(\frac{y}{\lambda + \|y\|_{\mathcal{H}}^2} \right) = y.$$

Hence

$$(\lambda I + y \otimes y)^{-1} y = \frac{y}{\lambda + \|y\|_{\mathcal{H}}^2},$$

including when $y = 0$. Thus,

$$\begin{aligned} \left\langle y, \widehat{A}_i^{-1} y \right\rangle_{\mathcal{H}} &\leq \left\langle y, (\lambda I + y \otimes y)^{-1} y \right\rangle_{\mathcal{H}} \\ &= \frac{\|y\|_{\mathcal{H}}^2}{\lambda + \|y\|_{\mathcal{H}}^2} \\ &\leq \frac{K}{\lambda + K}. \end{aligned}$$

The last inequality follows because $t \mapsto t/(\lambda + t)$ is increasing for $t \geq 0$.

Notice that y is an eigenvector of $y \otimes y$ when $y \neq 0$; we do not assume it is an eigenvector of $\widehat{T}_i$.

Step 4: Add the two terms. We obtain

$$\widehat{E}_i \leq \frac{K}{\lambda} + \frac{K}{\lambda + K}.$$

For the population result, repeat the argument using

$$\|\mu_Q\|_{\mathcal{H}}, \|\mu_i\|_{\mathcal{H}} \leq \sqrt{K}$$

and

$$T_i - \mu_i \otimes \mu_i = \mathbb{E}_{P_i}\left[(\phi(X) - \mu_i) \otimes (\phi(X) - \mu_i) \right] \succeq 0.$$

□

4 Null guarantee

Assume $Q = P_i$. For $\rho_0 \in (0,1)$, define

$$U_0 = \frac{K}{n\lambda}\left(1 + \sqrt{\log\frac{2}{\rho_0}}\right)^2, \tag{13}$$

$$a_0 = \sqrt{\frac{K}{2n}}\left(1 + \sqrt{\log\frac{2}{\rho_0}}\right), \tag{14}$$

$$H_0 = \frac{2[m_0 - a_0]_+^2}{K + \lambda}, \tag{15}$$

$$\eta_0 = \frac{U_0}{U_0 + H_0}. \tag{16}$$

Theorem 1 (Acceptance under the null)**.** *Under Assumptions 1 and 2,*

$$Q = P_i \quad \Longrightarrow \quad \mathbb{P}\{\widehat{S}_i \le \eta_0\} \ge 1 - \rho_0.$$

Moreover, $\eta_0 < 1$ *whenever* $m_0 > a_0$.

Proof. **Step 1: Expected mean-difference error.** Write $\mu = \mu_i$ and

$$F = \|\widehat{\mu}_Q - \widehat{\mu}_i\|_{\mathcal{H}}.$$

Under the null, independence and centering give

$$\begin{aligned}\mathbb{E}F^2 &= \mathbb{E}\,\|(\widehat{\mu}_Q - \mu) - (\widehat{\mu}_i - \mu)\|_{\mathcal{H}}^2 \\ &= \mathbb{E}\,\|\widehat{\mu}_Q - \mu\|_{\mathcal{H}}^2 + \mathbb{E}\,\|\widehat{\mu}_i - \mu\|_{\mathcal{H}}^2 \\ &= \frac{2}{n}\left(\mathbb{E}k(X,X) - \|\mu\|_{\mathcal{H}}^2\right) \le \frac{2K}{n}.\end{aligned}$$

Therefore,

$$\mathbb{E}F \le \sqrt{\frac{2K}{n}}.$$

Step 2: Concentration of the mean difference. Replacing one of the $2n$ observations changes F by at most $\sqrt{2K}/n$. The sum of squared bounded-difference constants is

$$2n\left(\frac{\sqrt{2K}}{n}\right)^2 = \frac{4K}{n}.$$

One-sided McDiarmid concentration yields

$$\mathbb{P}\{F - \mathbb{E}F \ge t\} \le \exp\left(-\frac{nt^2}{2K}\right).$$

Taking

$$t = \sqrt{\frac{2K}{n}\log\frac{2}{\rho_0}}$$

gives, with probability at least $1 - \rho_0/2$,

$$F \le \sqrt{\frac{2K}{n}}\left(1 + \sqrt{\log\frac{2}{\rho_0}}\right).$$

Hence

$$\widehat{D}_i \le \frac{F^2}{2\lambda} \le U_0.$$

Step 3: Concentration of the midpoint. Under the null, $\widehat{c}_i$ is the empirical mean of $2n$ iid observations from P_i. For

$$G = \|\widehat{c}_i - \mu\|_{\mathcal{H}},$$

we have

$$\mathbb{E}G \le \sqrt{\mathbb{E}G^2} \le \sqrt{\frac{K}{2n}}.$$

Replacing one observation changes G by at most $\sqrt{2K}/(2n)$, so the sum of squared constants is

$$2n\left(\frac{\sqrt{2K}}{2n}\right)^2 = \frac{K}{n}.$$

Therefore,

$$\mathbb{P}\{G - \mathbb{E}G \ge t\} \le \exp\left(-\frac{2nt^2}{K}\right).$$

Taking

$$t = \sqrt{\frac{K}{2n}\log\frac{2}{\rho_0}}$$

gives $G \le a_0$ with probability at least $1 - \rho_0/2$.

Step 4: Combine numerator and denominator bounds. By a union bound, both concentration events hold with probability at least $1 - \rho_0$. Their independence is not required.

On their intersection,

$$\|\widehat{c}_i\|_{\mathcal{H}} \ge [\|\mu\|_{\mathcal{H}} - a_0]_+ \ge [m_0 - a_0]_+.$$

Lemma 1 gives

$$\widehat{E}_i \ge \widehat{D}_i + H_0.$$

If $H_0 > 0$, then

$$\widehat{S}_i \le \frac{\widehat{D}_i}{\widehat{D}_i + H_0} \le \frac{U_0}{U_0 + H_0},$$

because $d \mapsto d/(d + H_0)$ is increasing. If $H_0 = 0$, then $\eta_0 = 1$ and the result follows from $\widehat{S}_i \le 1$. □

5 Standard separation guarantee

The standard proof controls the full mean-embedding errors and the operator error. It is conservative because the operator norm controls every feature-space direction, including directions unnecessary for a particular pair.

Assumption 3 (Pearson separation)**.** *For the fixed pair (Q, P_i),*

$$D_i(Q) \ge \delta > 0.$$

At the population level, Lemma 2 immediately gives

$$\boxed{S_i(Q) \ge \frac{\delta}{M_\lambda}.} \tag{17}$$

Indeed, $D_i(Q) > 0$ implies $E_i(Q) > 0$, and $E_i(Q) \le M_\lambda$.

5.1 Joint concentration

For $\rho_1 \in (0,1)$, define

$$b_n(\rho_1) = \sqrt{\frac{K}{n}}\left(1 + \sqrt{\log\frac{3}{\rho_1}}\right), \tag{18}$$

$$e_n(\rho_1) = \frac{K}{\sqrt{n}}\left(1 + \sqrt{2\log\frac{3}{\rho_1}}\right). \tag{19}$$

Lemma 3 (Mean and operator concentration)**.** *With probability at least* $1-\rho_1$*, all three inequalities hold:*

$$\|\widehat{\mu}_Q - \mu_Q\|_{\mathcal{H}} \le b_n(\rho_1), \qquad \|\widehat{\mu}_i - \mu_i\|_{\mathcal{H}} \le b_n(\rho_1),$$

and

$$\|\widehat{T}_i - T_i\|_{\mathrm{op}} \le e_n(\rho_1).$$

Proof. **Step 1: Concentration of one mean embedding.** For an iid empirical mean $\widehat{\mu}$ with population mean μ,

$$\mathbb{E}\,\|\widehat{\mu} - \mu\|_{\mathcal{H}}^2 = \frac{1}{n}\left(\mathbb{E}k(X,X) - \|\mu\|_{\mathcal{H}}^2\right) \le \frac{K}{n}.$$

Thus

$$\mathbb{E}\,\|\widehat{\mu} - \mu\|_{\mathcal{H}} \le \sqrt{\frac{K}{n}}.$$

Replacing one observation changes the norm error by at most $\sqrt{2K}/n$. McDiarmid gives

$$\mathbb{P}\left\{\|\widehat{\mu} - \mu\|_{\mathcal{H}} > \sqrt{\frac{K}{n}} + t\right\} \le \exp\left(-\frac{nt^2}{K}\right).$$

Take

$$t = \sqrt{\frac{K}{n}\log\frac{3}{\rho_1}}.$$

This proves each mean bound with failure probability at most $\rho_1/3$.

Step 2: Expected operator error. Let $V = \phi(X)\otimes\phi(X)$. In the Hilbert space of Hilbert–Schmidt operators,

$$\|V\|_{\mathrm{HS}} = \|\phi(X)\|_{\mathcal{H}}^2 \le K.$$

Independence and centering imply

$$\mathbb{E}\|\widehat{T}_i - T_i\|_{\mathrm{HS}}^2 = \frac{1}{n}\mathbb{E}\|V - \mathbb{E}V\|_{\mathrm{HS}}^2 \le \frac{K^2}{n}.$$

Consequently,

$$\mathbb{E}\|\widehat{T}_i - T_i\|_{\mathrm{HS}} \le \frac{K}{\sqrt{n}}.$$

Step 3: Concentration of the operator error. Replacing one observation changes the Hilbert–Schmidt norm error by at most

$$\frac{1}{n}\|\phi(x)\otimes\phi(x) - \phi(x')\otimes\phi(x')\|_{\mathrm{HS}} \le \frac{2K}{n}.$$

Thus,

$$\mathbb{P}\left\{\|\widehat{T}_i - T_i\|_{\mathrm{HS}} > \frac{K}{\sqrt{n}} + t\right\} \le \exp\left(-\frac{nt^2}{2K^2}\right).$$

Take

$$t = K\sqrt{\frac{2}{n}\log\frac{3}{\rho_1}}$$

and use

$$\|\widehat{T}_i - T_i\|_{\mathrm{op}} \le \|\widehat{T}_i - T_i\|_{\mathrm{HS}}.$$

This proves the operator bound with failure probability at most $\rho_1/3$.

Step 4: Combine the events. A union bound over the two mean bounds and the operator bound gives total failure probability at most ρ_1. □

5.2 Empirical score lower bound

Define

$$s = \sqrt{\frac{2}{\lambda}}\, b_n(\rho_1), \qquad r = \frac{e_n(\rho_1)}{\lambda},$$

and

$$L(\delta, \rho_1) = \frac{[\sqrt{\delta} - s]_+^2}{M_\lambda(1+r)}. \tag{20}$$

Theorem 2 (Alternative guarantee)**.** *Under Assumptions 1 and 3,*

$$\mathbb{P}\{\widehat{S}_i \ge L(\delta, \rho_1)\} \ge 1 - \rho_1.$$

Consequently, if $\eta < L(\delta, \rho_1)$, then

$$\mathbb{P}\{\widehat{S}_i > \eta\} \ge 1 - \rho_1.$$

Proof. Work on the event of Lemma 3.

Step 1: Bound the mean-difference error. Set

$$\widehat{\Delta}_i = \widehat{\mu}_Q - \widehat{\mu}_i.$$

The triangle inequality gives

$$\begin{aligned}\left\|\widehat{\Delta}_i - \Delta_i\right\|_{\mathcal{H}} &\le \|\widehat{\mu}_Q - \mu_Q\|_{\mathcal{H}} + \|\widehat{\mu}_i - \mu_i\|_{\mathcal{H}} \\ &\le 2b_n(\rho_1).\end{aligned}$$

Step 2: Compare empirical and population metrics. The operator error bound implies

$$\widehat{A}_i \preceq A_i + e_n(\rho_1) I.$$

Since $A_i \succeq \lambda I$,

$$e_n(\rho_1) I \preceq \frac{e_n(\rho_1)}{\lambda} A_i.$$

Hence

$$\widehat{A}_i \preceq (1+r) A_i.$$

Inverting gives

$$\widehat{A}_i^{-1} \succeq \frac{A_i^{-1}}{1+r}.$$

Step 3: Apply the reverse triangle inequality. Define

$$\|f\|_{A_i^{-1}} = \sqrt{\langle f, A_i^{-1} f\rangle_{\mathcal{H}}}.$$

Because $A_i^{-1} \preceq \lambda^{-1} I$,

$$\|\widehat{\Delta}_i - \Delta_i\|_{A_i^{-1}} \leq \frac{2b_n(\rho_1)}{\sqrt{\lambda}}.$$

Therefore,

$$\begin{aligned}\|\widehat{\Delta}_i\|_{A_i^{-1}} &\geq \left[\|\Delta_i\|_{A_i^{-1}} - \|\widehat{\Delta}_i - \Delta_i\|_{A_i^{-1}}\right]_+ \\ &\geq \left[\sqrt{2D_i(Q)} - \frac{2b_n(\rho_1)}{\sqrt{\lambda}}\right]_+.\end{aligned}$$

Step 4: Lower-bound empirical Pearson. Combining the previous inequalities,

$$\begin{aligned}\widehat{D}_i &\geq \frac{\|\widehat{\Delta}_i\|^2_{A_i^{-1}}}{2(1+r)} \\ &\geq \frac{[\sqrt{D_i(Q)} - s]_+^2}{1+r} \\ &\geq \frac{[\sqrt{\delta} - s]_+^2}{1+r}.\end{aligned}$$

Step 5: Normalize using the energy upper bound. If this lower bound is positive, then $\widehat{E}_i > 0$. Lemma 2 gives

$$\widehat{S}_i = \frac{\widehat{D}_i}{\widehat{E}_i} \geq \frac{\widehat{D}_i}{M_\lambda} \geq L(\delta, \rho_1).$$

If the bound is zero, the claim follows from $\widehat{S}_i \geq 0$, including the convention for $\widehat{E}_i = 0$. □

6 Nonempty threshold interval

Combining the null and standard alternative guarantees, a certified threshold satisfies

$$\boxed{\eta_0 \leq \eta < L(\delta, \rho_1).}$$

This interval is nonempty if and only if

$$\eta_0 < \frac{[\sqrt{\delta} - s]_+^2}{M_\lambda(1+r)}.$$

Since $\eta_0 > 0$, the positive part must be strictly positive. Multiplying by the denominator and taking square roots gives

$$\sqrt{\delta} - s > \sqrt{M_\lambda(1+r)\eta_0}.$$

Equivalently,

$$\boxed{\delta > \delta_{\text{crit}} := \left[s + \sqrt{M_\lambda(1+r)\eta_0}\right]^2.} \tag{21}$$

This is a necessary and sufficient condition for the displayed certified interval to be nonempty, not a necessary condition for successful classification.

For $\rho_0 = \rho_1 = \rho$, write

$$U = \frac{K}{n\lambda}\left(1 + \sqrt{\log\frac{2}{\rho}}\right)^2,$$

and

$$H = \frac{2}{K+\lambda}\left[m_0 - \sqrt{\frac{K}{2n}}\left(1 + \sqrt{\log\frac{2}{\rho}}\right)\right]_+^2.$$

Then

$$\eta_0 = \frac{U}{U+H},$$

and the explicit condition is

$$\boxed{\delta > \left[\sqrt{\frac{2K}{n\lambda}}\left(1+\sqrt{\log\frac{3}{\rho}}\right) + \sqrt{M_\lambda\left[1+\frac{K}{\lambda\sqrt{n}}\left(1+\sqrt{2\log\frac{3}{\rho}}\right)\right]\frac{U}{U+H}}\right]^2.}$$

The null proof combines two concentration events, explaining $\log(2/\rho)$. The standard alternative combines three, explaining $\log(3/\rho)$.

If $m_0 \le a_0$, then $\eta_0 = 1$ and no useful interval can be certified. For fixed positive m_0, K, λ, ρ, as $n \to \infty$,

$$U = O(n^{-1}), \qquad H \longrightarrow \frac{2m_0^2}{K+\lambda} > 0.$$

Consequently,

$$\eta_0 = O(n^{-1}), \qquad \delta_{\mathrm{crit}} = O(n^{-1}).$$

A Alternative directional proof

The standard proof controls errors in all feature-space directions. To certify separation of a fixed pair, one can instead control a single informative population direction. This gives smaller error terms without changing the score or its implementation.

Define

$$s_{\mathrm{dir}} = \sqrt{\frac{K}{n\lambda}\log\frac{2}{\rho_1}}, \qquad r_{\mathrm{dir}} = \frac{K}{\lambda}\sqrt{\frac{1}{2n}\log\frac{2}{\rho_1}},$$

and

$$L_{\mathrm{dir}}(\delta,\rho_1) = \frac{[\sqrt{\delta} - s_{\mathrm{dir}}]_+^2}{M_\lambda(1+r_{\mathrm{dir}})}. \tag{22}$$

Theorem 3 (Directional alternative guarantee)**.** *Under Assumptions 1 and 3,*

$$\mathbb{P}\{\widehat{S}_i \ge L_{\mathrm{dir}}(\delta,\rho_1)\} \ge 1-\rho_1.$$

Consequently, if $\eta < L_{\mathrm{dir}}(\delta,\rho_1)$*, then*

$$\mathbb{P}\{\widehat{S}_i > \eta\} \ge 1-\rho_1.$$

Proof. Write $D = D_i(Q)$, $A = A_i$, and $\Delta = \Delta_i$.

Step 1: Choose a fixed population direction. Since $D > 0$, define

$$u = \frac{A^{-1}\Delta}{\sqrt{2D}}.$$

This direction depends on the population distributions, not on the sampled windows. It is only a proof device; the classifier does not need to compute it.

Using $2D = \langle \Delta, A^{-1}\Delta\rangle_{\mathcal{H}}$ gives

$$\langle u,\Delta\rangle_{\mathcal{H}} = \sqrt{2D}, \qquad \langle u, Au\rangle_{\mathcal{H}} = 1.$$

Since $A \succeq \lambda I$,

$$1 = \langle u, Au \rangle_{\mathcal{H}} \geq \lambda \|u\|_{\mathcal{H}}^2, \qquad \|u\|_{\mathcal{H}}^2 \leq \frac{1}{\lambda}.$$

Step 2: Restrict the empirical variational problem. Define

$$\widehat{d} = \langle u, \widehat{\mu}_Q - \widehat{\mu}_i \rangle_{\mathcal{H}}, \qquad \widehat{q} = \left\langle u, \widehat{A}_i u \right\rangle_{\mathcal{H}}.$$

Restricting the empirical version of (2) to $f = tu$, $t \geq 0$, gives

$$\widehat{D}_i \geq \sup_{t \geq 0} \left(t\widehat{d} - \frac{t^2}{2}\widehat{q} \right).$$

Since $\widehat{q} > 0$, the maximizing value is $t = [\widehat{d}]_+ / \widehat{q}$. Hence

$$\widehat{D}_i \geq \frac{[\widehat{d}]_+^2}{2\widehat{q}}. \tag{23}$$

Step 3: Concentrate the directional mean. Let

$$h(x) = \langle u, \phi(x) \rangle_{\mathcal{H}}.$$

Using (5),

$$|h(x) - h(y)| \leq \|u\|_{\mathcal{H}} \|\phi(x) - \phi(y)\|_{\mathcal{H}} \leq \sqrt{\frac{2K}{\lambda}}.$$

Thus the range of h has length at most $\sqrt{2K/\lambda}$.

Moreover,

$$\widehat{d} = \frac{1}{n} \sum_j h(Z_j) - \frac{1}{n} \sum_j h(X_j), \qquad \mathbb{E}\widehat{d} = \sqrt{2D}.$$

The sum of squared range lengths of the $2n$ independent summands is at most

$$2n \left(\frac{1}{n} \sqrt{\frac{2K}{\lambda}} \right)^2 = \frac{4K}{n\lambda}.$$

One-sided Hoeffding concentration gives

$$\mathbb{P}\{\widehat{d} < \sqrt{2D} - t\} \leq \exp\left(-\frac{n\lambda t^2}{2K} \right).$$

Taking $t = \sqrt{2}\, s_{\mathrm{dir}}$ yields

$$\widehat{d} \geq \sqrt{2D} - \sqrt{2}\, s_{\mathrm{dir}}$$

with probability at least $1 - \rho_1/2$.

Step 4: Concentrate the directional quadratic term. We have

$$\widehat{q} = \frac{1}{n} \sum_j h(X_j)^2 + \lambda \|u\|_{\mathcal{H}}^2, \qquad \mathbb{E}\widehat{q} = \langle u, Au \rangle_{\mathcal{H}} = 1.$$

Also,

$$0 \leq h(X_j)^2 \leq \|u\|_{\mathcal{H}}^2 \|\phi(X_j)\|_{\mathcal{H}}^2 \leq \frac{K}{\lambda}.$$

Hoeffding therefore gives

$$\mathbb{P}\{\widehat{q} > 1 + t\} \leq \exp\left(-\frac{2n\lambda^2 t^2}{K^2} \right).$$

Taking $t = r_{\mathrm{dir}}$ yields

$$\widehat{q} \le 1 + r_{\mathrm{dir}}$$

with probability at least $1 - \rho_1/2$.

Step 5: Combine the two scalar bounds. By a union bound, both events hold with probability at least $1 - \rho_1$. Their independence is not required. On their intersection, (23) gives

$$\begin{aligned}\widehat{D}_i &\ge \frac{[\sqrt{2D} - \sqrt{2}\, s_{\mathrm{dir}}]_+^2}{2(1 + r_{\mathrm{dir}})} \\ &= \frac{[\sqrt{D} - s_{\mathrm{dir}}]_+^2}{1 + r_{\mathrm{dir}}} \\ &\ge \frac{[\sqrt{\delta} - s_{\mathrm{dir}}]_+^2}{1 + r_{\mathrm{dir}}}.\end{aligned}$$

Step 6: Normalize. When this bound is positive, $\widehat{E}_i > 0$ and Lemma 2 gives

$$\widehat{S}_i \ge \frac{\widehat{D}_i}{M_\lambda} \ge L_{\mathrm{dir}}(\delta, \rho_1).$$

When the bound is zero, the claim follows from $\widehat{S}_i \ge 0$. □

A.1 Threshold condition and comparison

The null guarantee is unchanged. The directional certified interval is

$$\eta_0 \le \eta < L_{\mathrm{dir}}(\delta, \rho_1).$$

As in the standard analysis, it is nonempty precisely when

$$\boxed{\delta > \delta_{\mathrm{crit,dir}} := \left[s_{\mathrm{dir}} + \sqrt{M_\lambda(1 + r_{\mathrm{dir}})\eta_0}\right]^2.} \tag{24}$$

For the same parameters,

$$s_{\mathrm{dir}} = \sqrt{\frac{K}{n\lambda}}\sqrt{\log\frac{2}{\rho_1}},$$

whereas

$$s = \sqrt{\frac{2K}{n\lambda}}\left(1 + \sqrt{\log\frac{3}{\rho_1}}\right).$$

Likewise,

$$r_{\mathrm{dir}} = \frac{K}{\lambda\sqrt{n}}\sqrt{\frac{1}{2}\log\frac{2}{\rho_1}},$$

whereas

$$r = \frac{K}{\lambda\sqrt{n}}\left(1 + \sqrt{2\log\frac{3}{\rho_1}}\right).$$

Therefore,

$$s_{\mathrm{dir}} < s, \qquad r_{\mathrm{dir}} < r.$$

It follows that

$$L_{\mathrm{dir}}(\delta, \rho_1) \ge L(\delta, \rho_1), \qquad \delta_{\mathrm{crit,dir}} < \delta_{\mathrm{crit}}.$$

The lower-bound inequality is strict whenever $L_{\mathrm{dir}}(\delta, \rho_1) > 0$.